



Semester 1 Examinations 2016/2017

Exam Code(s) 1SD3
Exam(s) Higher Diploma in Software Design and Development

Module Code(s) CT537

Module(s) Software Engineering Methods

Paper No. 1
Repeat Paper

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Instructions: Candidates should answer **Question 1** and any **two other** questions.
All questions carry equal marks.

Duration 2 hours

No. of Pages 4

Requirements:

None

Release to Library: Yes

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No

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- 1.(a) Draw a **context diagram**, a **systems level DFD** and one **third level DFD** to represent the data and processing involved in the following system description of a travel agency:

An Irish-based travel agent called **Adventure Breaks** specialises in offering tailor-made adventure holidays. It now wishes to offer its services to online customers. Therefore a website is being developed that will allow customers to pre-book an adventure holiday for a specific period. The website will provide the following functionality at a minimum:

- A customer will be asked to enter their personal details, select preferred destination and dates and provide details of the number of persons travelling.
- If the customer has a discount code they can enter it at this stage; if they have travelled with Adventure Breaks before, they can include the details.
Note that returning customers are to be offered a 5% discount on holiday packages.
- The system displays a series of holiday packages ordered by increasing price (includes flight and hotel details).
- The user can then select the most preferred package and request any further information they require.
- The customer can either pay a deposit or pay in full (electronically) for it at this time.
- On receipt of the customer's deposit or payment, the customer is emailed an order confirmation number.
- The holiday must be paid for in full at least two weeks before the departure date.

The website will also facilitate the cancellation or modification of a booking.

- The system will allow a user to cancel or modify a booking by entering their order confirmation number. The policy of **Adventure Breaks** is to allow the modification or full cancellation of an order only if it is made more than one week before the pickup time.

(16)

- (b) In structured analysis and design, what is the relationship between the ERD and DFD models? Illustrate your answer using examples from the *Adventure Breaks* system explored in Q.1(a) above.

(4)

2. (a) *Mobile* systems have some unique characteristics which differentiate them from more traditional systems, and require a different approach to be applied in their specification, design and development. Prepare a **testing strategy** or plan for a *Mobile* system of your choice. Your strategy should incorporate both low and high order testing, and should be illustrated with practical examples of testing techniques where appropriate.

(9)

(b) Provide a brief report detailing how the **implementation** of the *Adventure Breaks* system in Q.1 will be approached. The report should detail the design method, programming language and framework employed, as well as the implementation strategy stating all assumptions made.

(5)

(c) In assessing the quality of a proposed design, software designers examine the coupling, cohesion and overall factoring of their solution. What are the guidelines which exist for the designer in this regard?

Identify the types of coupling evident in each of the following interfaces from the calling module to a called module that calculates NET PAY:

- (a) CALL CALCULATE NET PAY USING GROSS PAY, TAX CREDIT, TAX RATE
- (b) CALL CALCULATE NET PAY USING PAY_DETAILS
- (c) CALL CALCULATE NET PAY USING VALID_FLAG, PAY_DETAILS

(6)

3. (a) Explain why the intangibility of software systems poses special problems for software project management.

(3)

(b) Suppose you are managing a project which is getting behind schedule. Possible actions include renegotiating the time schedule, adding people to the project, and renegotiating quality requirements. In which ways can these actions shorten the time schedule? What other actions could you take to finish the project on time?

(5)

(c) Cost estimates are inherently risky, irrespective of the estimation technique used. How would you reduce the risks inherent in a cost estimate?

(4)

(d) Prepare a **requirements search strategy** for a new Customer Relationship Management (CRM) system for an international distribution company. Clearly indicate the stakeholders and the requirements gathering techniques you have chosen, ensuring you have considered identification of both functional and non-functional requirements in your outlined strategy.

(8)

4. (a) Select an appropriate **development process** for the **Adventure Breaks** system described in Q.1 and outline the reasons for your selection. In your answer consider the strengths and weaknesses of different software processes and the types of projects and circumstances to which they are suited.

(6)

(b) What is a structured **process specification**? What are its objectives? Select an appropriate **process specification technique** for the following bank sub-system and use it to describe the business logic represented:

A bank uses the following rules to classify new accounts:

If a depositor's age is 21 or above and if the deposit is €100 or more, classify the account as type A.

If the depositor is under 21 and the deposit is €100 or more, classify it as type B.

If the depositor is 21 or over and the deposit is below €100 classify it as type C.

If the depositor is under 21 and the deposit is below €100 then do not open account.

(9)

(c) **Software maintenance** activities can be classified as corrective (fixing errors), adaptive (responding to change) and perfective (improving the original software). Discuss why this distinction is useful and how you would expect the proportion of time spent on each activity to change as the software ages.

(5)